MATH 2400 - Midterm 2

14.1: Functions of Several Variables

This is all pretty straight-forward. Things like domain and range follow from the properties of the functions under consideration extended from one dimension to two or three. The best thing to do is graph regions in the xy-plane when dealing with domain questions for $f(x, y)$.

The main new-ish idea in this section are **level curves** and **contour plots**. A level curve is the curve you get in the xy-plane when you set $f(x, y) = c$. A contour plot is just multiple level curves plotted on the xy-plane simultaneously.

14.2: Limits and Continuity

Limits are much more complicated for multivariable functions because now you have to take into consideration the infinitely many different paths along which a point can be approached, instead of just the two one-sided limits that exist when working with functions of a single variable. For this reason, we usually evaluate limits to show that they **DON'T** exist by choosing two different paths (usually something simple like approaching along the x or y axis) and showing that you get two different limiting values. There are some situations where you can do algebra or use the Squeeze Theorem to show that a limit exists.

Continuity is nicer in that basically anything/everything that's continuous as a single variable function is also continuous as a multi variable function. Polynomials, Exponentials, basic Trig, Log's, etc, are a all continuous at all points where they are defined. Rational functions are continuous except where the denominator is 0. Things go through pretty much that same here.

14.3: Partial Derivatives

Things start to get interesting here. In general, finding a partial derivative is no harder than finding the derivative of a single variable functions: just treat one of the variables as a constant and differentiate with respect to the other variable. Things to keep in mind:

1. There are two first-order partial derivatives for $f(x, y)$: $f_x(x, y), f_y(x, y)$. These will almost always be different.
2. There are four second-order partial derivatives of $f(x, y)$: $f_{xx}(x, y), f_{yy}(x, y), f_{xy}(x, y), f_{yx}(x, y)$.
3. If the mixed second-order partials are continuous at all points in a disk, then $f_{xy}(x, y) = f_{yx}(x, y)$ for all (x, y) inside that disk. This is formally known as Clairaut's Theorem and infomrally known as 'mixed partials commute.'
4. Partial derivatives give you the slope of tangent lines in planes parallel to the x and y-coordinate axes.
5. All of this generalizes in a natural way to $f(x, y, z)$ and functions of even more independent variables. There are as many first-order partials as there are independent variables, but the number of higher-order derivatives get complicated.

14.4: Tangent Planes

For a function $z = f(x, y)$ the best linear approximation to f at the point (x_0, y_0) is given by the tangent plane to the surface at $(x_0, y_0, f(x_0, y_0))$.

When you have an explicit function $z = f(x, y)$, the tangent plane is given by:

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Notice that this a generalization of the expression for the tangent line.

If a surface is defined parametrically as $\vec{r}(u,v)$, then the tangent plane at the point $\vec{r}(u_0,v_0)$ is given by

$$(\vec{r}_u(u_0,v_0)\times\vec{r}_v(u_0,v_0))\cdot(x-x_0,y-y_0,z-z_0)=0$$

where $(x_0,y_0,z_0)=\vec{r}(u_0,v_0)$.

14.5: The Chain Rule

Diagrams are **KEY** here.

Starting with the dependent variable at the top, draw branches to all of the intermediate variables, continuing until you get to the last set of independent variables. On each branch, differentiate the variable at the top with respect to the variable at the bottom, using a partial derivative if there is more than one branch connected to the top variable, an ordinary derivative if there is only one branch connected to the top variable. To find the derivative with respect to one of the base-level independent variables, find all branches that end in that variable, trace up the chain, multiplying all of the derivatives together, and add all the pieces that you get from all of the branches that end in the variable you are differentiating with respect to.

Something to keep in mind: intermediate variables should occur in pairs, once in the numerator and once in the denominator. If you 'cancel' these out (which you don't ever actually do) you should be left with the dependent variable in the numerator and the independent variable in the denominator.

This is all much easier to understand if you **DRAW A DIAGRAM!**

14.6: Directional Derivatives and the Gradient Vector

Given $f(x,y)$ the **gradient** of f, written ∇f is equal to the vector of partial derivatives:

$$\nabla f(x,y)=\langle f_x(x,y),f_y(x,y)\rangle$$

For $f(x,y,z)$ you will have a vector with three partial derivatives.

The **directional derivative** of $f(x,y)$ at the point (x_0,y_0) in the direction of $\vec{u}=\langle a,b\rangle$ is given by

$$D_{\vec{u}}f(x_0,y_0)=\nabla f(x_0,y_0)\cdot\vec{u}$$

Please note that $\vec{u}$ must be a **unit vector**. If you want to find the directional derivative in the direction of a non-zero, non-unit vector $\vec{v}$, you must first normalize it: $\vec{u}=\frac{\vec{v}}{|\vec{v}|}$.

The directional derivative gives the slope of the tangent line to the surface given by $f(x,y)$ at the point $(x_0,y_0,f(x_0,y_0))$ in the plane that passes through the point and is parallel to $\vec{u}$.

The gradient has some very nice geometric properties. They are:

1. $\nabla f(x_0,y_0)$ points in the direction of greatest increase.
2. $|\nabla f(x_0,y_0)|$ is the rate of greatest increase.
3. $-\nabla f(x_0,y_0)$ points in the direction of greatest decrease, and $|-\nabla f(x_0,y_0)|$ is the magnitude of greatest decrease.
4. If $F(x,y,z)=c$ defines a level surface, then $\nabla f(x_0,y_0,z_0)$ is normal to the level surface at any point $P=(x_0,y_0,z_0)$ on that surface. This means that the tangent plane to $F(x,y,z)=k$ at the point P can be written

$$\nabla F(x_0,y_0,z_0)\cdot(x-x_0,y-y_0,z-z_0)=0$$

5. The same reasoning as in the previous item holds if you have $F(x,y)=k$ giving a level curve: $\nabla F(x_0,y_0)$ is normal to the level curve at the point $P=(x_0,y_0)$ on the curve.

14.7: Maximum and Minimum Values

Just like in Calc 1, you use first-order (partial) derivatives to find critical points and second-order (partial) derivatives to classify them as maxs or mins.

To find critical points, find all values (a, b) such that $f_x(a, b) = 0$ and $f_y(a, b) = 0$ (or where the first-order partial derivatives don't exist). Once you have the critical points, use the Second Derivative Test to classify them.

Start by forming $D = D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - (f_{xy}(a, b))^2$. Then

1. If $D > 0$ and $f_{xx}(a, b) > 0$ then $f(a, b)$ is a local minimum.
2. If $D > 0$ and $f_{xx}(a, b) < 0$ then $f(a, b)$ is a local maximum.
3. If $D < 0$ then $f(a, b)$ is a saddle point (and so neither a local max nor local min).
4. If $D = 0$, then the second derivative test is inconclusive and doesn't tell you anything about the point (a, b).

14.8: Lagrange Multipliers

To find the maximum and minimum values of $f(x, y)$ subjected to the constraint $g(x, y) = k$ (assuming that $\nabla g \neq \vec{0}$ on the level curve $g(x, y) = k$)

1. Find all values of x, y, λ such that $\nabla f(x, y) = \lambda \nabla g(x, y)$ and $g(x, y) = k$.
2. Evaluate f at all points found in the previous step. The largest of these are the maximum values and the smallest are the minimum values of f.

All of this goes through exactly the same when you are dealing with $f(x, y, z)$ and the constraining level surface $g(x, y, z) = k$. Just add $z's$ to the functions in the previous steps.

15.1: Double Integrals over Rectangular Regions

In order to evaluate the double integral of $f(x, y)$ over the rectangular region $R = [a, b] \times [c, d]$, you will use **iterated integrals**:

$$\int\int_R f(x, y)dA = \int_a^b \int_c^d f(x, y)dydx = \int_c^d \int_a^b f(x, y) = dxdy$$

You **MUST** pay attention to the order of integration: the limits of integration must match with the order of the dx, dy. Over a rectangular region R, you can interchange the order of integration by just switching the limits of integration and the dx, dy.

Depending on $f(x, y)$, one order will be much more difficult (if not impossible) than the other. Look at the structure of the function and try to determine which order will make things easier, and then go with it. If you start down a path and things become very complicated (or impossible), start over using the other order.

In the special case that $f(x, y) = g(x)h(y)$ you can 'separate' the two integrals:

$$\int\int_R g(x)h(y)dA = \left(\int_a^b g(x)dx\right)\left(\int_c^d h(y)dy\right)$$

15.2: Double Integrals over General Regions

If a general region D can be defined by $a \leq x \leq b$ and $g_1(x) \leq y \leq g_2(x)$, then

$$\int\int_D f(x, y)dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)dydx$$

If a general region D can be defined by $c \leq y \leq d$ and $h_1(y) \leq x \leq h_2(y)$, then

$$\int\int_D f(x,y)dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)dxdy$$

PLEASE NOTE: the limits of the outer integrals must always be constants, while the limits of the inner integral can be functions of the outer variable.

For many 'simple' regions D (think triangles and things like that) you can change the order of integration. This will involve understanding the boundary of D as functions of x and y and thinking about filling D with vertical line segments (if you want the order to be $dydx$) or horizontal line segments (if you want the order to be $dxdy$). The non-vertical and non-horizontal portions of the boundary of D must be invertible and you need to make sure that you are expressing the boundary in terms of the correct independent variable (following the note above). The easiest thing to do is draw the region D in the xy-plane and think about drawing vertical or horizontal line segments (you're focusing on the values of the starting and stopping points of the line segments).

In general, given any 'simple' region D, the area can always be calculated as a double integral:

$$\text{Area}(D) = \int\int_D 1 \, dA$$

Okay, it probably seems like a lot when you see it all in one place. And to a certain extent it **IS** a lot. But, almost everything on this midterm has a direct connection to things you've done before in either Calc 1 or Calc 2. How do you find a partial derivative? Differentiate using all the rules that you learned in Calc 1. How do you evaluate a double integral? Integrate using all the techniques that you learned in Calc 2. So while it is a lot, there's not really all that much that is truly **new**.

The most computationally confusing material on this midterm (the Chain Rule and Lagrange Multipliers) can be simplified enormously by using diagrams. **USE THEM!**

Don't try to visualize everything all at once. That's too hard and is usually unnecessary. For example, when calculating the volume of a solid given by a function $f(x,y)$ over some region D, you really don't need to know what the graph of $z = f(x,y)$ looks like. What you need to understand is the region D, which will live in the xy-plane and which you should be able to sketch. Once you have something visual for this, get your limits of integration, think about the appropriate order of integration, and just start integrating.

While on the topic of integration, you have to find the Goldilocks Zone where you're not trying to do too much (like evaluating an integral that is impossible to evaluate) while also not trying to do too little (like not integrating correctly). This is a skill that takes time to develop. If you trust your integration, then if things look like they're getting messy, maybe you should try a different order of integration. If it's less messy, keep going. But it's important that you not make mistakes and create the mess through that.

Remember, at the end of the day, all of these integrals are **DEFINITE INTEGRALS** so if your answer isn't a number, you're doing something wrong.

Also remember that we're only going to ask you to do things that we think you can actually do. If you're asked to evaluate an integral, you should be able to evaluate it without too much strain or stress (assuming of course that you're not making mistakes). If you keep running into a dead end, either try something different or take a long, slow, careful look at what you're doing and see if there are any obvious mistakes. If both orders of integration seem impossible, then you are definitely either making a mistake somewhere (a sin of commission) or you are overlooking something (a sin of omission).

You've got to know how to differentiate (all of the different derivatives that we've introduced: first-order partials, second-order partials, chain rule, directional derivatives) and how to use these derivatives. Sometimes it's as simple as getting a number and interpreting it as a rate of change. Other times, you're finding critical points or maxs and mins. Still other times, you're finding a tangent plane. Pretty much all of these ideas have analogues in Calc 1, so don't think that you need to reinvent the wheel here.

And always remember what they say about getting to Carnegie Hall: Practice, Practice, Practice.

But don't forget the counterpart to that maxim: practice doesn't make perfect; perfect practice makes perfect.